Rohit Kumar

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Skills

- **Languages**: Python, C/C++, Java, Kotin
- **Frameworks**: HTML and CSS.
- **Tools/Platforms**: Ms Excel, MySQL, Git & GitHub
- **Soft Skills:** Problem - Solving, Adaptability

Projects

- **Movie Recommendation System** | GitHub — Nov' 25 – Dec' 25
  - Developed a movie recommendation system using unsupervised machine learning techniques to suggest Movies based on user preferences and similarity patterns to enhance personalized user recommendations.
  - Implemented clustering and similarity-based algorithms to group movies/users without labeled data.
  - Used feature extraction (genre, ratings, keywords) to represent movies numerically for machine learning.
  - Applied cosine similarity / distance metrics to generate personalized recommendations.
  - Designed the system to provide content-based recommendations efficiently.
  - Improved recommendation accuracy by optimizing similarity calculations and feature selection.
  - **Tech: Python, Machine Learning, Unsupervised Learning, Pandas , NumPy.**

- **Mental Health Simulation System:**| GitHub — Jul' 25 – Aug '25
  - Designed and developed a console-based mental health simulation system using C++ to model user mental states and behavioral responses based on predefined rules and dynamic user inputs.
  - Implemented Data Structures such as arrays, stacks, queues, linked lists, and maps to manage user data, mood history, and session tracking with efficient storage, faster access, and structured data flow mechanisms.
  - Applied DSA concepts like searching, sorting, and recursion for efficient data retrieval and analysis of mental health patterns to support predictive insights, trend identification, emotional correlations.
  - Simulated stress, anxiety, and mood level changes based on user inputs and predefined rules.
  - Developed logic for risk assessment and recommendations using conditional algorithms and state transitions.
  - Ensured modular and optimized code structure using OOP concepts (classes, objects, encapsulation).
  - Improved program performance by optimizing time and space complexity of core operations.
  - **Tech: C++, DSA, STL, OOPs**

- **Deadlock Detection System** | GitHub — Apr' 25 – May '25
  - Developed a Deadlock Detection System to identify deadlock conditions in an operating system environment .
  - Implemented deadlock detection algorithms using resource allocation graphs and process-resource mapping for efficient dependency tracking for deadlock identification and system analysis.
  - Analyzed circular wait conditions to detect deadlocks among multiple processes execution and synchronization.
  - Tracked resource allocation, requests, and availability using suitable data structures for optimization.
  - Gained hands-on experience with operating system concepts such as deadlocks, synchronization, and resource management.
  - **Tech: HTML, CSS, Javascript**

Cpp with DSA & OOPs | Certificate — Jun '25 – Jul '25
- About: The hands-on training in C++ programming focused on Object-Oriented Programming (OOPs) and Data Structures & Algorithms.
- Gained practical experience in classes, objects, inheritance, polymorphism, abstraction, and encapsulation.
- Implemented arrays, linked lists, stacks, and queues, along with searching and sorting algorithms.
- Enhanced proficiency in C++ syntax, STL , memory management, and clean modular code design.

Certificates

- Java Programming Certification – Neo Colab | Link — May '25
- Master Generative AI & Generative AI Tools | Infosys Springboard | Link

Education

- **Lovely Professional University** — Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 5.71*** — *Aug 2023 – Present.*
- **Laxmi Narayan Dubey College** — Bihar, India
  Intermediate; ***Percentage: 75%*** — *Jun 2021 - Mar 2021*
- **Noneya High School** — Bihar, India
  Matriculation; ***Percentage: 82%*** — *Apr 2018 - Apr 2019*